

PoseHelper: ML-Powered Human Pose Reference Tool for Artists

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PROBLEM

Artists constantly need human pose references when drawing figures — but searching Google or Pinterest returns cluttered photographs that are hard to trace and can unintentionally influence drawing style. There is no widely accessible tool that converts a photo into a clean, minimal skeleton reference an artist can draw from directly.

SOLUTION

PoseHelper lets artists upload any single-person photo and instantly receive a clean 2D skeleton mannequin as a drawing reference, along with a predicted pose category label they can use to search for additional references.

ML PIPELINE

Stage 1 — Keypoint Detection

MoveNet Thunder (TensorFlow Hub) extracts 17 body landmarks from the uploaded photo, producing a 34-dimensional feature vector (x, y per keypoint).

```
TensorFlow version: 2.20.0
MoveNet model loaded

Pose detected! 17 keypoints:
00 nose           : x=0.483, y=0.249, confidence=0.605
01 left_eye       : x=0.487, y=0.231, confidence=0.724
02 right_eye      : x=0.482, y=0.235, confidence=0.698
03 left_ear       : x=0.521, y=0.204, confidence=0.729
04 right_ear      : x=0.506, y=0.215, confidence=0.482
05 left_shoulder  : x=0.596, y=0.221, confidence=0.856
06 right_shoulder : x=0.482, y=0.275, confidence=0.715
07 left_elbow     : x=0.667, y=0.156, confidence=0.700
08 right_elbow    : x=0.391, y=0.292, confidence=0.760
09 left_wrist     : x=0.654, y=0.071, confidence=0.768
10 right_wrist    : x=0.420, y=0.206, confidence=0.554
11 left_hip       : x=0.535, y=0.484, confidence=0.869
12 right_hip      : x=0.456, y=0.477, confidence=0.659
13 left_knee      : x=0.562, y=0.664, confidence=0.883
14 right_knee     : x=0.425, y=0.654, confidence=0.875
15 left_ankle     : x=0.628, y=0.840, confidence=0.673
16 right_ankle    : x=0.403, y=0.840, confidence=0.739
```

Stage 2 — Pose Classification

A trained sklearn MLP classifier predicts one of 5 pose categories: Downdog, Goddess, Plank, Tree, Warrior2. Three models compared: Logistic Regression, KNN, and MLP

```
Loaded 1080 samples, 52 columns
Class distribution:
label      count
plank      266
warrior2   252
downdog    222
goddess    180
tree       160
Name: count, dtype: int64

Train: 864 samples | Test: 216 samples

Logistic Regression    → Accuracy: 85.2%
K-Nearest Neighbors    → Accuracy: 81.9%
MLP Classifier         → Accuracy: 85.2%

Best model: Logistic Regression (85.2%)
```

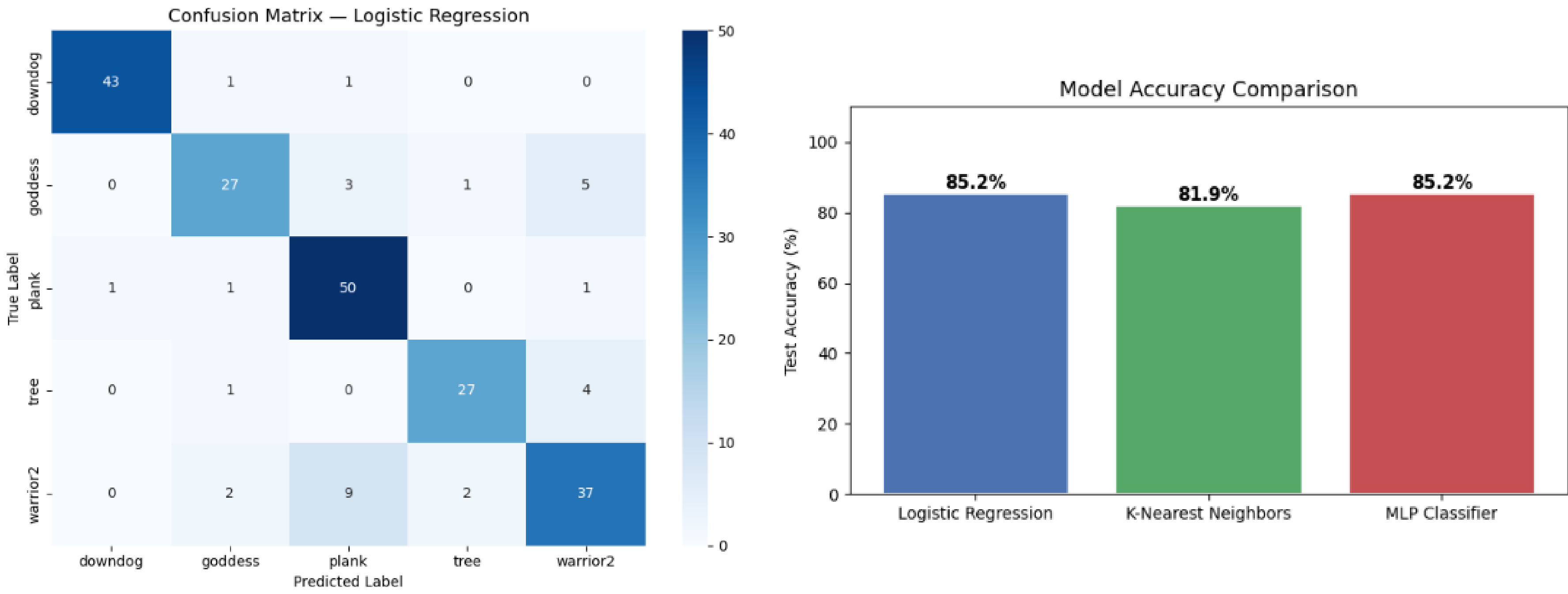
Stage 3 — Mannequin Rendering

OpenCV maps keypoints onto a blank canvas — limbs as colored lines, joints as circles, head as a proportional filled circle. Output: downloadable PNG reference.

DATASET & TRAINING

- Yoga Poses Dataset (Kaggle) — 1,080 labeled images across 5 classes
- Keypoints extracted via MoveNet, saved as CSV (34 features + label)
- 80/20 stratified train/test split · StandardScaler normalization

RESULTS



Best models: Logistic Regression and MLP tied at 85.2% · Macro F1: 0.85
Strongest: Downdog (F1: 0.97) · Tree (F1: 0.87)
Most challenging: Warrior2 (F1: 0.76) — most confused with Plank (9 misclassifications)

KEY TAKEAWAY

PoseHelper demonstrates that a lightweight ML pipeline — pretrained keypoint detection + a trained classifier + rule-based rendering — can solve a real creative workflow problem. The minimalist skeleton gives artists maximum freedom: it captures pose geometry without imposing style, body type, or character — making it ideal for artists who want pose reference without creative constraint.

A full-body action illustration that typically takes 4–5 hours was completed in under 2 hours using PoseHelper as a reference.

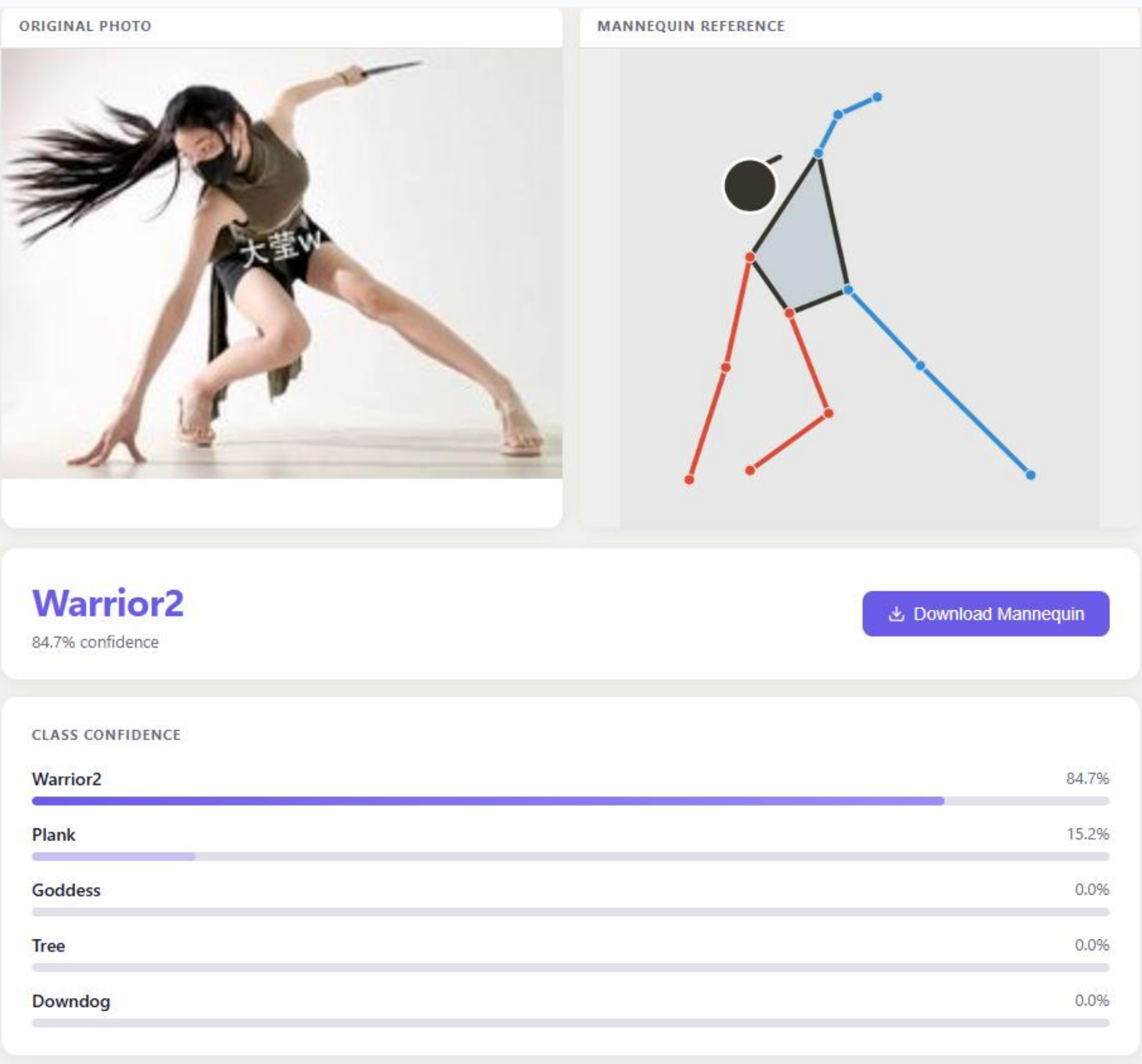
Live App: huggingface.co/spaces/OspreyMoby/posehelper
GitHub: github.com/Ruiyang1210W/posehelper

LIMITATIONS & FUTURE WORK

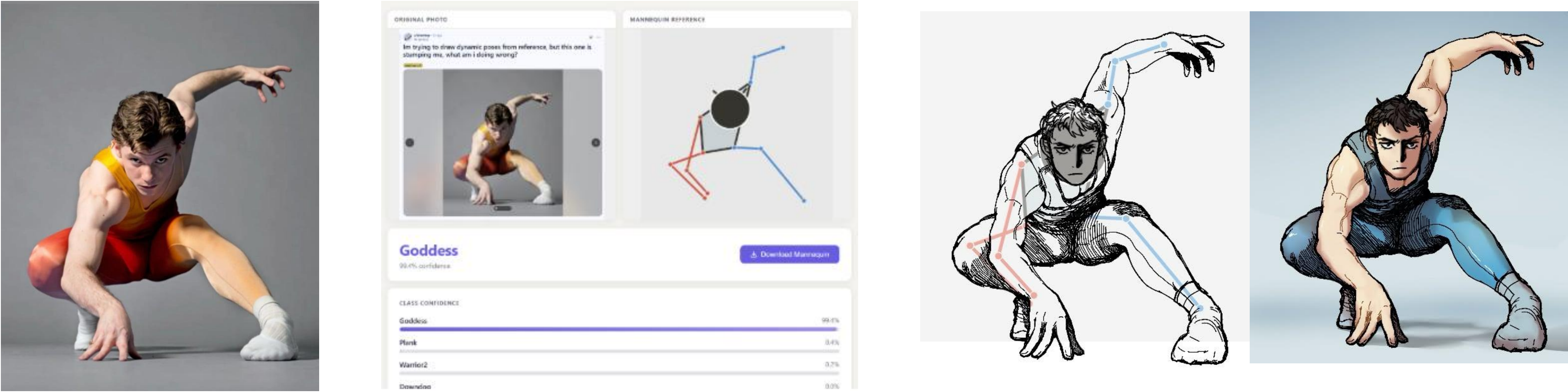
- Currently supports single-person images only (MoveNet SinglePose)
- Future: MoveNet MultiPose support · expanded pose categories beyond yoga · adjustable mannequin styles

Reference:

Bazarevsky, V., et al. (2020). *BlazePose: On-device Real-time Body Pose Tracking*. arXiv:2006.10204.
Cao, Z., et al. (2017). *Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields*. CVPR 2017.
Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
Lin, T.Y., et al. (2014). *Microsoft COCO: Common Objects in Context*. ECCV 2014.
Dataset:
Yoga Poses Dataset: Niharika (2021). *Yoga Poses Dataset*. Kaggle. kaggle.com/datasets/niharika41298/yoga-poses-dataset



DEMO



Original Photo.
Save as JPG, PNG, WEBP

Upload the photo → get a clean skeleton reference instantly. The predicted pose label doubles as a search term, helping artists find additional references by pose category rather than describing it manually.

Sketch over the skeleton, using the joint positions to lock in accurate proportions. Refine your lines, then add color, shading, and final details. The result: a full-body, full-color illustration with a dynamic pose — in a fraction of the usual time.

